

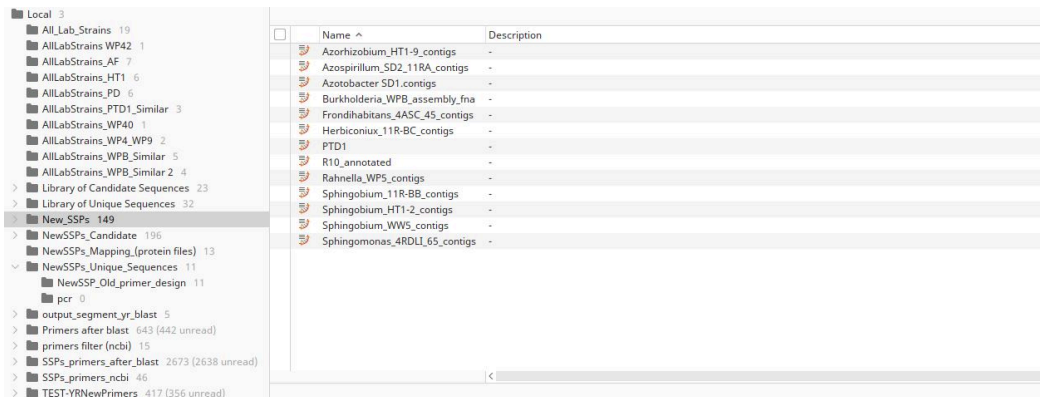
Strain Specific Primers (SSPs) Protocol

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1. Download and Organize Genomes:

- Download the DNA genome files.
- Place those genomes in a designated folder named after 'Your_project/strains'.



Genome Folder Named After Project

2. Segment Sequences to 500 Base Pairs:

- Utilize the segment_script to segment the sequences to 500 base pairs (requires downloading 'shred' from bbmap, for information on how to download, scroll to download bbmap).
- Note: When using the segment_script make sure to have no spaces between the file name. The code will not work if there are spaces. Additionally, make sure the file ends with .fasta.

▼ Code

```
#!/bin/bash

assembly_dir="path/to/your/script"
output_dir="path/to/your/script"
shred_path="path/to/your/bbmap/shred.sh"

mkdir -p "$output_dir"

echo "Processing files in $assembly_dir directory:"
for file in "$assembly_dir"/*.fasta; do
    if [[ -f "$file" ]]; then
        echo "Processing file: $file"
        genome=$(basename "$file" .fasta)
        segmented_file="${output_dir}/${genome}-segmented.fasta"
```

```

"$shred_path" in="$file" out="$segmented_file" length=500

total_reads=$(grep -c '^>' "$file")

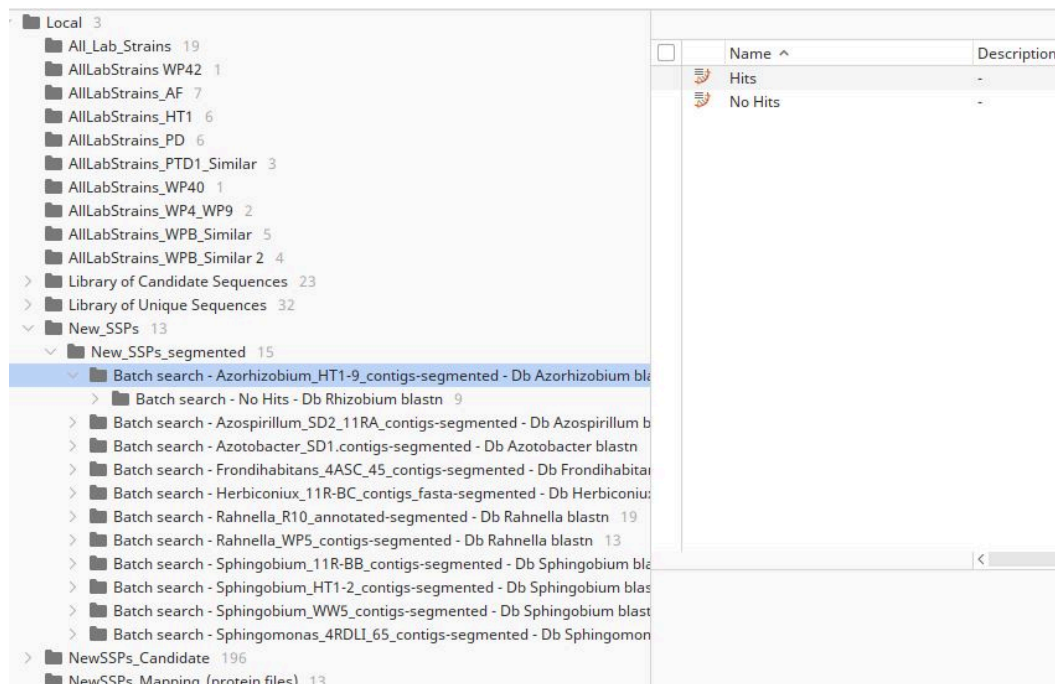
start_bp=1
end_bp=500

echo ">Contig_1 $total_reads reads from $genome (trimmed) (normalized) assembled
fi
done

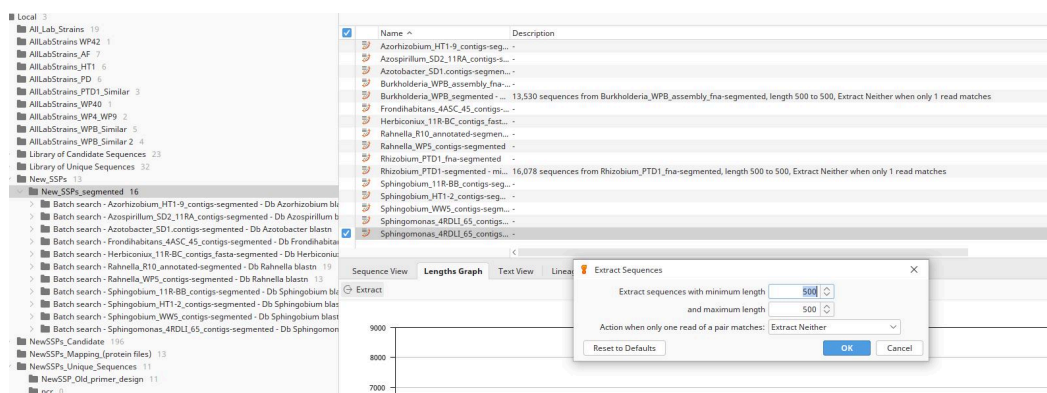
echo "Script execution completed."

```

- Save the segmented genome files in a folder named '{Your_project/strains}_segmented'.
- Make sure to delete contigs or sequences less than 500 base pairs. If not, it can affect designing primers. Select file > Length Graph > Extract > Extract Length Min = 500, Max = 500, Neither > Ok.



Segmented Sequences Subfolder



Deleting Contigs With Less Than 500 Base Pairs

3. Download Databases:

- Use the 'download metadata' code to fetch the NCBI RefSeq database.
- Note: For the 'download metadata' code, the NCBI refseq link should be replaced every month as NCBI constantly updates their database, scroll down to more information for help on how to update.

▼ Code

```
#!/bin/bash

curl -o assembly_summary.txt ftp://ftp.ncbi.nlm.nih.gov/genomes/refseq/assembly_summa
```

- Use the 'filter_and_download' code to download databases for your genera and related ones for blasting.
- Note: For the 'filter_and_download' code, make sure to correctly spell the name of the genus, or the code will not work. Additionally, the code is not key sensitive, typing Dyella and dyella, should return the similar and correct files.

▼ Code

```
#!/bin/bash

database_dir="db_genus_NewSSP_Autumn"

mkdir -p "$database_dir"

echo "Enter the genus name:"
read genus

genus=$(echo "$genus" | tr '[:upper:]' '[:lower:]')

filtered_lines=$(awk -F '\t' -v OFS='\t' -v g="$genus" '

    BEGIN {

        print "## See ftp://ftp.ncbi.nlm.nih.gov/genomes/README_assembly_summary.txt for
        print "#assembly_accession\tbioproject\tbiosample\ttwgs_master\trefseq_category\tt

    }

    $12 == "Complete Genome" && tolower($8) ~ g {

        print

    }

')
```

```

' assembly_summary.txt)

filename="${genus}_assembly_summary.txt"
echo "$filtered_lines" > "$filename"

num_lines=$(wc -l < "$filename")

if [ "$num_lines" -gt 25 ]; then # Change 25 to the desired number
sampled_lines=$(shuf -n 25 "$filename") # Change 25 to the desired number
else
sampled_lines="$filtered_lines"
fi

sampled_filename="${genus}_sampled_summary.txt"
echo "$sampled_lines" > "$sampled_filename"

echo "Downloading fasta files..."
echo "What do you want to call this database?"
    read DB
    awk -F '$\t' '{print $20 "/" $1 "_" $16 "_cds_from_genomic.fna.gz"}' ${genus}_sampled

    wget -i assemblies.txt
    cat *.fna.gz >> db_${DB}.fasta.gz
    rm *.fna.gz

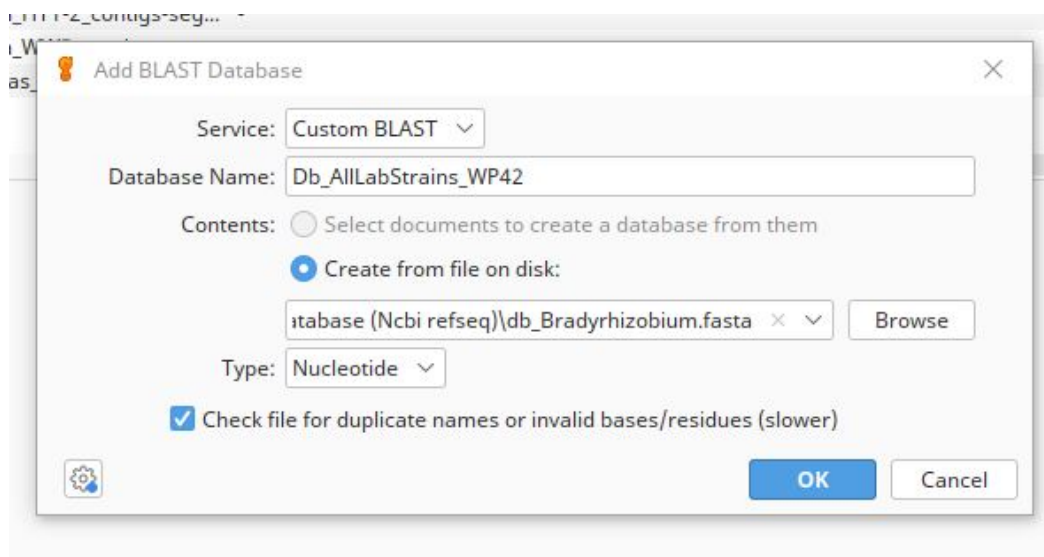
    gzip -dk db_${DB}.fasta.gz
    mkdir db_${DB}_meta
    mv db_${DB}.fasta.gz db_${DB}_meta/
    mv _sampled_summary.txt db_${DB}_meta/db_${DB}_records.txt
    mv assemblies.txt db_${DB}_meta/db_${DB}_assemblies.txt

echo "Assembly files downloaded for $genus"

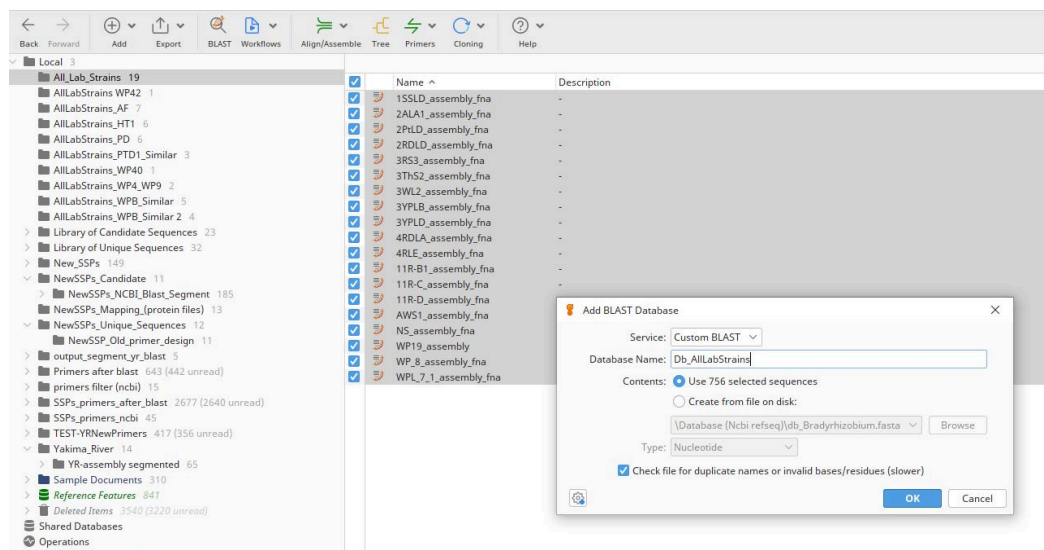
```

4. Create and Use Custom Blast Databases:

- Open Geneious Prime, Navigate to Tools > Add/Remove Databases.
- Add a custom blast database > Create from file on disk > Type: Nucleotide.
- Alternatively, upload databases manually in Geneious prime. Create a designated folder > Drag and drop files > Select all > Navigate to Tools > Add/Remove Databases > Select documents to create a database > Create a database name > Ok.



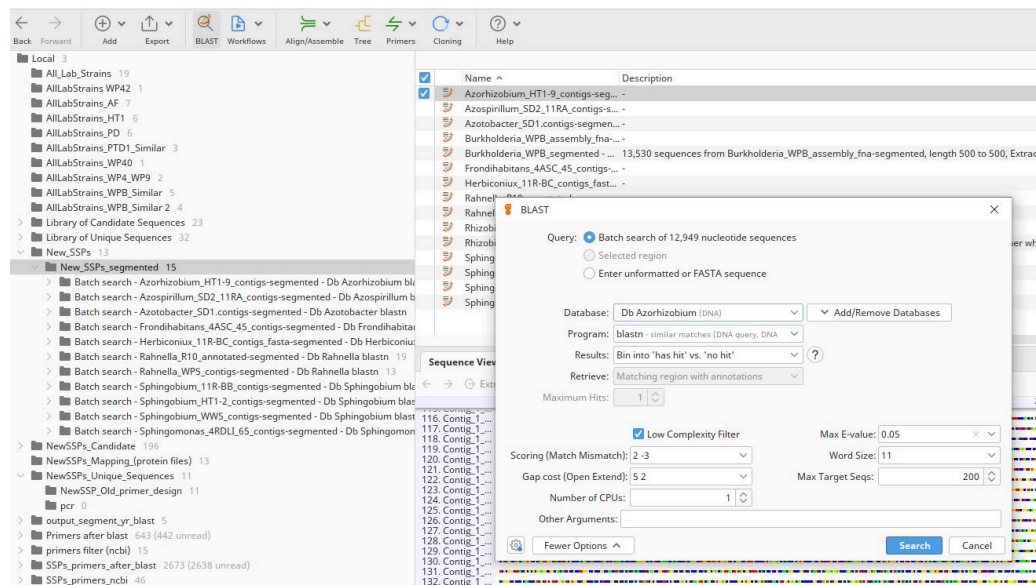
Add Custom Blast Database Downloaded Using Code



Add Custom Local Database

5. Blast Segmented Genome:

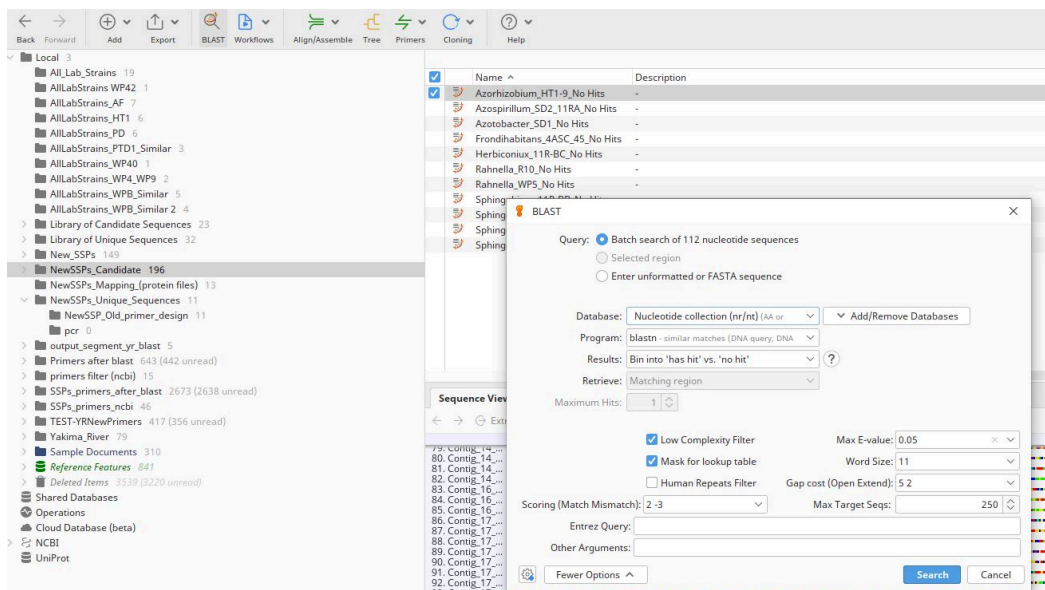
- Blast the segmented genome against its database and related ones (e.g., blast rhizobium against rhizobium and neorhizobium).
- Aim for a relatively small and unique number of sequences (≤ 130).
- Blast > Query: batch search > Click database > Select custom database > Program: blastn > Results: bin into "has hit" vs "no hit." > Search (Reference image below).
- Maximum hits are your preferences, however aim for around ≤ 250 .
- After blasting, place the 'no hits' into a folder named 'Library of candidate sequences'.



Blasting Segmented Genome Against Its Genus

6. Library of candidate sequences:

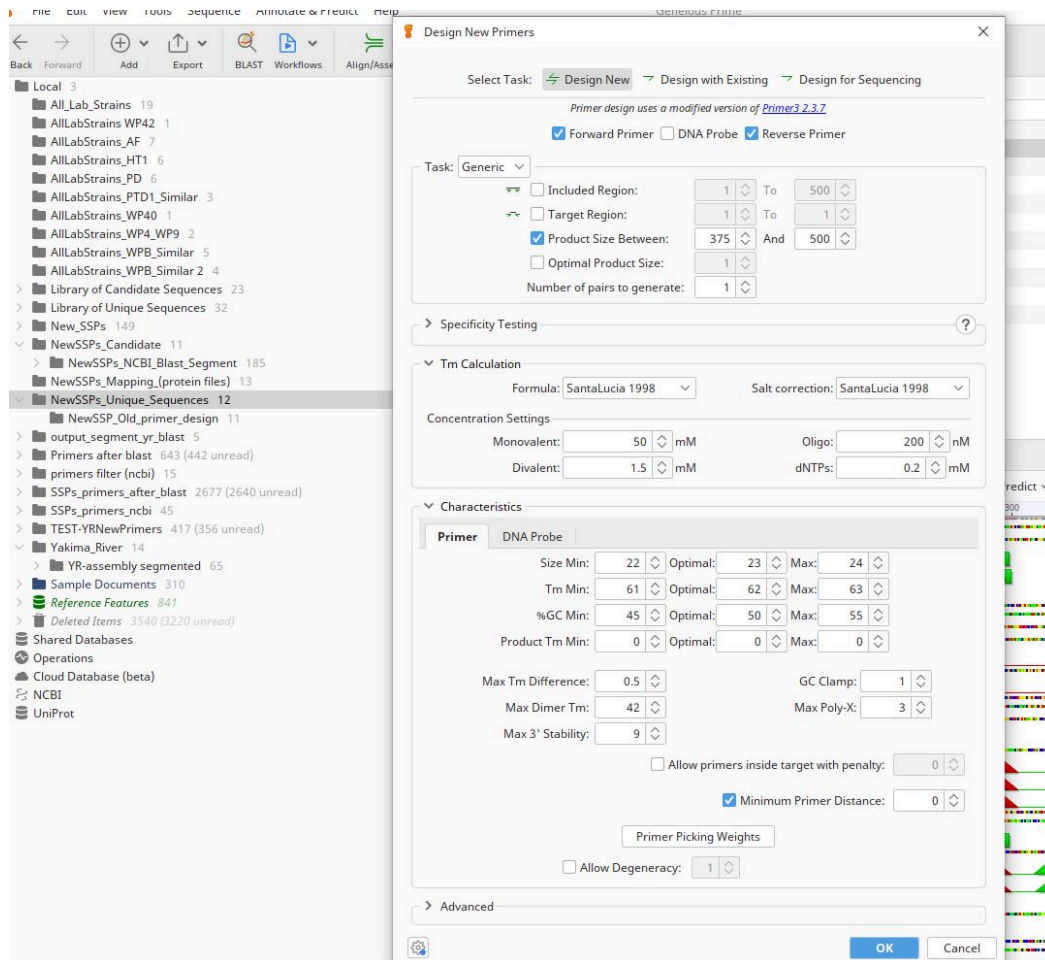
- Blast the 'no hits' against the NCBI database to receive the most unique sequences.
- Blast > Query: batch search > Database: nucleotide collection (nr/nt) > Program: blastn > Results: bin into "has hit" vs "no hit" > Search.
- Note: Make sure to rename your 'no hits' files using strain's name and the step. example : 'Your_strain_candidate'.
- Note: Blast one file at a time, if not it can return 'all hits' due to a lot of computing usage, scroll down to more information for help.



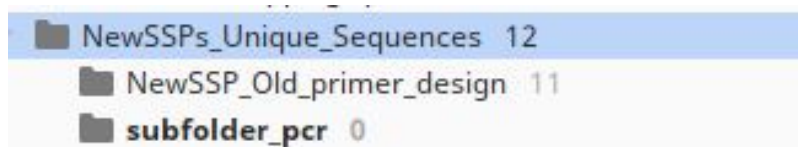
NCBI Blast For Library Of Candidate Sequences

7. Library of unique sequences:

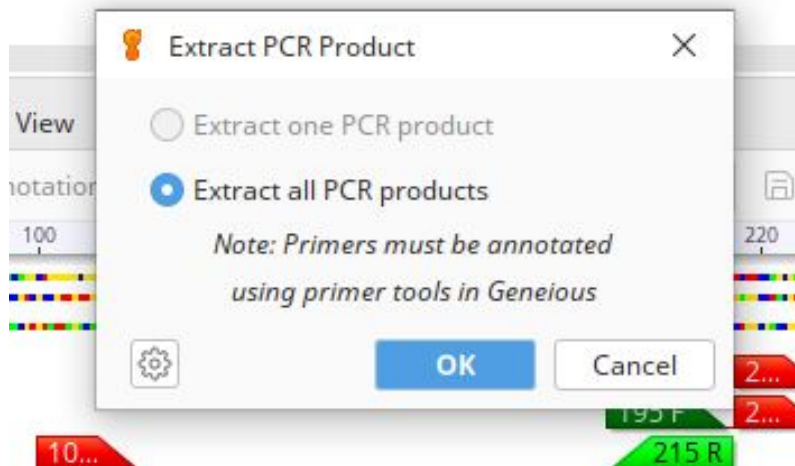
- After blasting against NCBI, place the 'no hits' into a folder named 'Library of unique sequences'.
- Design your SSP primers based on your preferences. Primers > Design new primers.
- After primers are created, make a sub-folder in the library of unique sequences. This folder will simplify the process of extracting PCR products.
- Once the file is in the folder, Click primers > Extract PCR product > All products > Ok.
- Once extracted, Right click on the products > Select group sequences, Name it after your strain.
- Note: Its best to do this file by file to make sure the PCR products don't mix with the wrong file.



Designing Primers



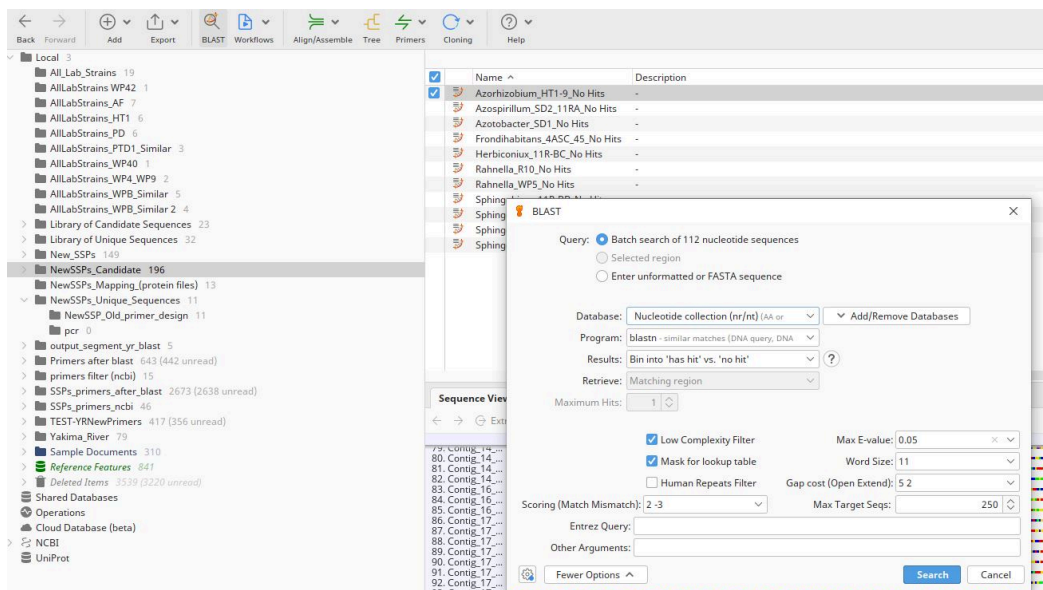
Sub-Folder



PCR Extraction

8. Blast primers against NCBI:

- Place the extracted PCR product files in a folder named "project_name_primer_blast".
- Once placed in the folder blast the primer file to ensure the primers are all unique.
- Blast > Query: batch search > Database: Nucleotide collection (nr/nt) > Program: blastn > Results: bin into "has hit" vs "no hit."
- Note: Blast one file at a time or it will return 'all hits' and not work, scroll down to more information for help.

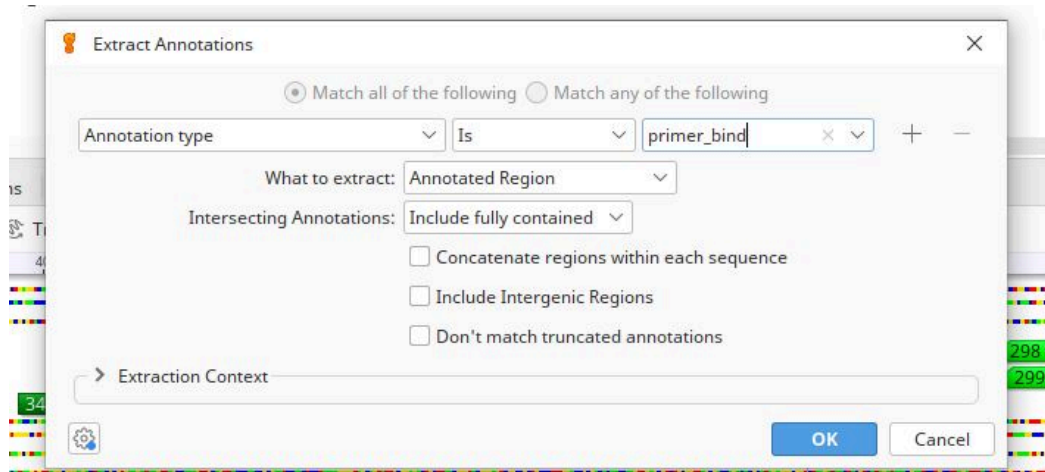


Blast Primers Against NCBI Parameters

9. Extract primer annotations:

- After blasting, place the 'no hits' into a folder named 'project_name_primer_extract'.
- Create a sub-folder named after the strain. Do this for all of the strains.

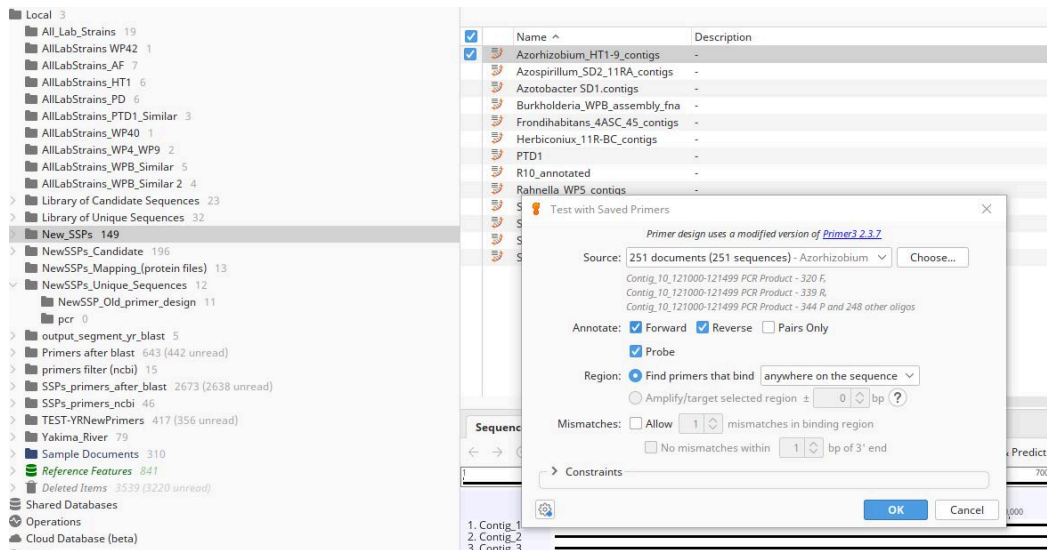
- Place the strains file in their corresponding sub-folders.
- Click on the file > Tools (navigation bar on top) > Extract annotations > Select primer binds > Ok.
- Note: If you have created primers with DNA probes, make sure to do the step above again and extract DNA probes instead of primer binds.
- After extracting, place the unique file back into the folder.



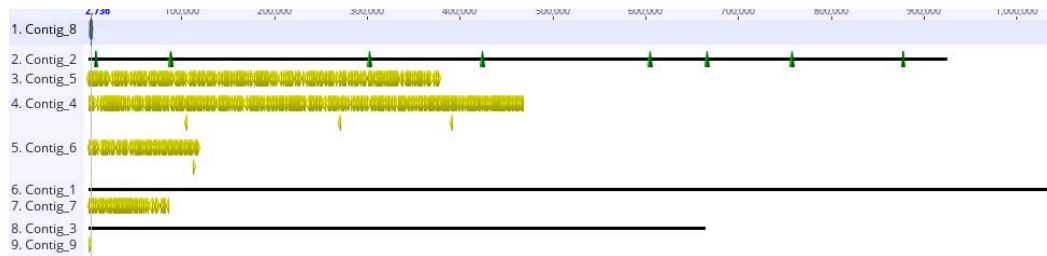
Extract Primer Annotations

10. Map primers to original genome:

- To map primers, go to the original genome files and select the file you want to map against.
- Then Select primers > Test with saved primers > Source: choose the primers that correspond to the reference sequence > Select Forward and Reverse > Ok.
- Note: Make sure you map the primers to its correct reference sequence. Ex. Azorhizobium primers should be mapped to Azorhizobium genome.
- When the primers are mapped correctly it should resemble this.



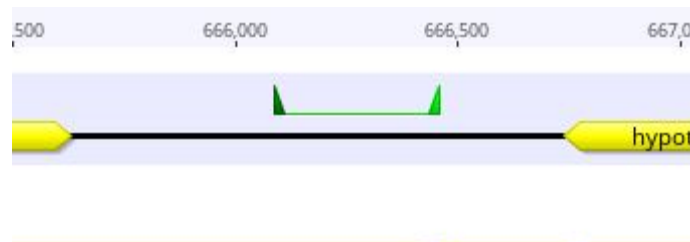
Mapping Primers With Test With Saved Primers



Correctly Mapped Primers

11. Rules for selecting good primers:

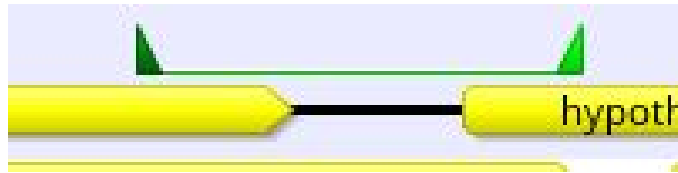
- The most unique primers should fall within regions. It is in order from best to least.
- Both primers should fall in non-coding region (CDS).
- One primer falls in non-CDS, other primer fall in CDS.
- Primers span across two-genes.
- Hypothetical proteins >> known genes.
- Once the best primers are selected, place them in a genomic data table containing information such as forward/reverse primer, forward/reverse sequence, forward/reverse length, %GC, Tm, contig and product size.



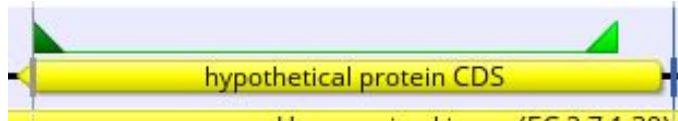
Best Primers - Fall on non cds



Second Best Primers - One on non cds



Third Best Primers - Spans Two Genes



Last Primer - On Hypothetical Gene

Downloading bbmap:

More information:

- Make sure to change the directory or path stated in the code, to where the code is located in your computer. Additionally, you can change the desired amount of segmenting or downloaded files in the scripts.
- Sometimes if multiple operations are running on Geneious it may return 'all hits' (no unique sequences) for NCBI blasts. This may happen due to many reasons.
- One reason is because there are too many operations happening and many people are using NCBI. To fix this, limit the number of operations you are doing on Geneious while blasting against NCBI.
 - An alternative is using NCBI. Use their blast and manually search for those contigs and place them in a file. When using NCBI, segment the files even smaller as it can take too much CPU power, use the script `segment_geneious` and change the segment number to 25.

Handling No Hits Against NCBI:

- If all sequences are 'no hits' when blasting against NCBI, consider segmenting the files.
- Use the 'segment_geneious' code* to segment the file into 25-contig files.

▼ Code

```
#!/bin/bash

if [ $# -lt 3 ]; then
    echo "Usage: $0 <input_fna_file> <output_prefix> <contigs_per_file>"
    exit 1
fi
```

```

input_file=$1
output_prefix=$2
contigs_per_file=25 # Change 25 to the desired number

if [ ! -f "$input_file" ]; then
    echo "Input FASTA file not found: $input_file"
    exit 1
fi

if [ "$contigs_per_file" -lt 1 ]; then
    echo "Invalid number of contigs per file: $contigs_per_file"
    exit 1
fi

input_filename=$(basename "$input_file")
output_dir="${output_prefix}_${input_filename%.*}_segments"
mkdir -p "$output_dir"

total_contigs=$(grep -c "^>" "$input_file")

num_segments=$(( (total_contigs + contigs_per_file - 1) / contigs_per_file ))

function split_file {
    input_file="$1"
    output_dir="$2"
    contigs_per_file="$3"
    segment_num="$4"

    awk -v start="^>" \
        -v end="^>" \
        -v contigs="$contigs_per_file" \
        -v segment="$segment_num" \
        'BEGIN {count = 0; file_num = 1}
        {
            if ($0 ~ start) {
                count++;
                if (count > contigs) {
                    count = 1;
                    file_num++;
                }
                if (file_num == segment) {
                    printf "%s\n", $0;
                    next;
                }
            }
            if (file_num == segment) {
                printf "%s\n", $0;
            }
        }
        {' "$input_file" > "${output_dir}/${output_prefix}_segment_${segment_num}.fna"
}

```

```
for (( segment_num = 1; segment_num <= num_segments; segment_num++ )); do
    split_file "$input_file" "$output_dir" "$contigs_per_file" "$segment_num"
done
```

```
echo "Segmentation completed successfully. Segmented files are stored in: $output_dir"
```

- Alternatively, use NCBI to blast. Click NCBI for the website > Select nucleotide blast > Database: standard > Nucleotide collection (nr/nt) > Program: blastn > Algorithm parameters should resemble geneious prime blast parameters. Now upload your segmented file (segmented to 25), name the file and click blast.
- To identify 'no hits' in NCBI it is light gray contigs with an asterik*. This option causes you to manually search for contigs in the files based on no hits.

The screenshot displays the NCBI BLASTN web interface. At the top, there are tabs for different BLAST programs: **blastn**, blastp, blastx, tblastn, and tblastx. The main heading is "BLASTN programs search nucleotide databases using a nucleotide query".

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) Query subrange [?](#)

From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus

New ☐ Experimental databases [Try experimental taxonomic nt databases](#) [Download](#)

For more info see [What are taxonomic nt databases?](#)

Nucleotide collection (nr/nt) [?](#)

Organism ☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to ☐ Sequences from type material

Entrez Query [?](#) [YouTube](#) [Create custom database](#)

Program Selection

Optimize for ☐ Highly similar sequences (megablast) ☐ More dissimilar sequences (discontiguous megablast) ☒ Somewhat similar sequences (blastn)

Choose a BLAST algorithm [?](#)

BLAST Search database nt using Blastn (Optimize for somewhat similar sequences)

☐ Show results in a new window

+ Algorithm parameters

NCBI Blast

Algorithm parameters

General Parameters

Max target sequences100
Select the maximum number of aligned sequences to display
Short queriesAutomatically adjust parameters for short input sequences
Expect threshold0.05
Word size11
Max matches in a query range0

Scoring Parameters

Match/Mismatch Scores2,-3
Gap CostsExistence: 5 Extension: 2

Filters and Masking

FilterLow complexity regions
Species-specific repeats for: Homo sapiens (Human)
MaskMask for lookup table only
Mask lower case letters

BLAST
Search database nt using Blastn (Optimize for somewhat similar sequences)
Show results in a new window

NCBI Blast Algorithm Parameters

Unix Help:

- Below are Unix/Terminal Commands used to call the scripts.
 - 'download_metadata': ./download_metadata.sh
 - 'filter_and_download': ./filter_and_download.sh
 - 'segmentScript': ./segmentScripts.sh
 - 'segment_geneious': ./'file name'.sh ~/location of file you want to segment/name of file you want to segment create a name for the segmented file #of contigs per file.
 - ex: ./geneious_segment.sh ~/Documents/Rahnella_R10_NCBI_Segment.fna Rahnella_R10_NCBI_Trial 25
- [Unix Tutorial](#)
- [Bioinformatics and Unix Tutorials](#)

Resources

- [Geneious](#)
- [Geneious Academy](#)
- [Geneious User Manual](#)
- [Geneious Prime Manual](#)

- [Geneious Support](#)
- [NCBI Blast](#)